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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 08/30/2023 |
| **PI Name:** | Gilbert, Jack Ph.D. | | |
| **Project Title:** | Quantifying Selection for Pathogenicity and Antibiotic Resistance in Bacteria and Fungi on the ISS – a Microbial Tracking Study | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | gilbertjack@gmail.com | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 630-915-2383 |
| **Organization Name:** | University of California, San Diego | | |
| **PI Address 1:** | 9500 Gilman Dr | | |
| **PI Address 2:** | MC 0763 | | |
| **PI Web Page:** | | | |
| **City:** | La Jolla | **State:** | CA |
| **Zip Code:** | 92093-5004 | **Congressional District:** | 52 |
| **Comments:** | | | |
| **Project Type:** | Flight,New Investigation | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research |
| **Start Date:** | 10/01/2019 | **End Date:** | 12/31/2023 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Griko, Yuri | **Contact Phone:** | 650-604-0519 |
| **Contact Email:** | Yuri.V.Griko@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2023 per NSSC information (Ed., 9/15/23)<br>NOTE: End date changed to 09/30/2023 per PI and NSSC information (Ed., 9/21/22) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Venkateswaran, Kasthuri Ph.D. ( NASA Jet Propulsion Laboratory ) | | |
| **Grant/Contract No.:** | 80NSSC19K1604 | | |
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| **Task Description:** | The built environment (BE) is a perfect model to examine microbial adaptation on the edge of survival. However, the role of the BE in mediating microbial selection remains unquantified. The International Space Station (ISS) provides a unique environment to study microbial dynamics, because it is a sealed system with a standing occupant population of approximately six people and visits spaced by 12-week periods. Notably, this time frame correlates with our previous work, which showed that bacteria acquire antibiotic resistance genes within 6 weeks in a hospital environment. This project will provide insights into the mechanisms of bacterial and fungal ecology, as it pertains to the development of antimicrobial resistance (AMR) and virulence phenotypes in the unique environment of the ISS. By isolating organisms that survive in these environments, quantifying their phenotypes in vitro and in vivo, sequencing their genomes, and leveraging metagenome-based genome reconstruction and ancestral genotype prediction, it should be possible to identify the genetic traits that code those microbial phenotypes (e.g. antibiotic resistance and virulence) that negatively influence human health. In this project, we are testing the hypothesis that built environments select for stress tolerance in human-associated microbes, which result in an increase in (i) genes and pathways associated with antibiotic resistance and pathogenicity, (ii) purifying selection of these traits, (iii) minimum inhibitory antibiotic concentration, and (iv) in vivo pathogenicity of isolated cultures in ISS microbes. We aim to examine this phenomenon in samples collected over a 20-year period that will leverage both the ISS Microbial Tracking database and a new series of collections.<br>Objective 1: To determine the distribution and selection of antimicrobial resistance and virulence genes in bacterial and fungal genomes and metagenomes in the ISS environment. Experiments have been designed to understand genes and pathways that are under purifying selection in the ISS environment. Previous NASA investigations of ISS environmental surfaces have isolated nearly 500 microorganisms, which are archived at Life Science Data Archive (LSDA) for future investigations. Strains were identified using partial 16S rRNA gene sequencing analysis, and hence, taxa were not fully characterized. Furthermore, an additional ~500 strains were collected in Microbial Tracking (MT) projects (MT-1; 202 strains and MT-2; 300 strains) and were phylogenetically identified using the full-length 16S rRNA gene sequence or Internal Transcribed Spacer (ITS) region. As these datasets lack in-depth genomic sequencing data, our ability to perform phylogenomic analysis to predict ancestral genotypes or genes and pathways that are under purifying selection in the ISS environment has been limited. This study leverages these two existing NASA culture collections, whereby strains are selected for analysis based on phylogeny, pathogenicity, etc. Selected isolates are being genome sequenced and the associated shotgun metagenomic sequencing is being mined. Tasks include:<br><br>Task 1. Sequence whole genomes of ISS microorganisms Task 2. Generate bioinformatic antibiotic resistance profiles of ISS microorganisms Task 3. Elucidate genes associated with host-invasion potential of ISS microorganisms Task 4. Assemble genomes from metagenomic data Task 5. Predict ancestral genotypes of key bacterial species<br><br>Objective 2: To quantify virulence and antimicrobial resistance phenotypes in archived ISS microbial isolates. While there is substantial evidence to suggest increased antibiotic resistance and virulence in space-flown isolates when compared to ground-based organisms, the extent to which the presence of resistance or virulence genes associates with the phenotype in vivo is unclear. We will determine how microbial genotypes, identified in Aim 1, impact AMR and virulence. Up to 200 microbial strains from LSDA, MT-1 and MT-2 collections will be cultured and subjected to detailed antibiotic resistance assays. We will also select 50 strains in which to validate and quantify virulence traits in vivo in nematode mortality models, which our preliminary data suggest are highly sensitive in detecting strain-specific virulence. The results of these assays will be correlated with genetic evidence of antibiotic resistance genes and virulence pathways. Whereas genes predicting resistance or virulence that fail to provide the predicted phenotype will be considered a “false positive”, genotypes that do not predict resistance or virulence will be considered a “false negative”. False negative resistance and virulence genes present potential new therapeutic targets. Tasks include:<br><br>Task 6. Perform in vitro antibiotic resistance assays Task 7. Perform in vivo virulence assays using C. elegans<br><br>Objective 3: To utilize live-dead cell characterization methods to determine the distribution and temporal dynamics of living microorganisms, as well as antimicrobial resistance and virulence phenotype in the ISS environment. Less than 30% of intact, potentially viable microorganisms isolated from previous MT flights could be cultured, which has made comprehensive characterization of bacterial and fungal species, and potentially critical genomic changes to AMR and virulence genes, impossible to discern. We have previously developed and deployed propidium monoazide (PMA) treatment to allow molecular characterization of AMR and virulence genes in living cells, which we have shown correlates with phenotypic susceptibility and WGS data of cultured isolates. By applying metagenomics to PMA-treated samples, we plan to capture genotype assemblies from a broader phylogenetic diversity than can be retrieved from culturing alone. In objective three, we have planned a new flight series to use PMA-based molecular analysis, in combination with traditional methods. Phylogenetic analysis, metagenomic sequencing, and phenotypic assays will be performed as in Objectives 1 and 2, and changes in AMR and virulence will be compared to MT-1 and MT-2 missions. MT-1, MT-2, and new data collected will be compiled for analysis to generate a model of AMR and virulence dynamics in the ISS. Tasks include:<br><br>Task 8. ISS environmental sampling Task 9. Estimate and identify culture-based microbial burden and diversity Task 10. Perform metagenomic sequencing and qPCR of PMA-treated DNA to examine the viable phylogenetic and functional diversity |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Built ecosystem-associated microbes are predominantly sourced from the occupants, which is restricted to humans on the ISS, we can elucidate the contribution and consequences of environment-driven genomic changes that will be critical for the future design and operation of terrestrial built environments in ways that promote ‘healthy’ microbial ecosystems. Further, Long term, identification of microbial taxa that pose a particular threat to crew health can be targeted to develop antimicrobial treatment strategies to protect not only crew members from ISS-specific pathogens, but also to tackle earth-bound antibiotic resistance, which has become a world-wide health crisis. |

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| **Task Progress:** | During the fourth year of our award, we completed all antibiotic resistance assays and C. elegans-based virulence experiments for MT-1, -2, and -3 missions (Tasks 6 and 7).<br>qPCR assays and DNA extractions were completed for all MT-3 bacterial and fungal isolates (n = 1113), and isolate samples have been submitted for whole genome sequencing (Task 10). Additionally, we completed all MT-3 metagenomic sequencing and have begun assembly and annotation of those data for subsequent quantitative and comparative analyses (Task 10). With this, we have now obtained all existing community-level data required for project completion.<br><br>Ed. Note:<br><br>Tasks include -<br><br>Task 1. Sequence whole genomes of ISS microorganisms<br><br>Task 2. Generate bioinformatic antibiotic resistance profiles of ISS microorganisms<br><br>Task 3. Elucidate genes associated with host-invasion potential of ISS microorganisms<br><br>Task 4. Assemble genomes from metagenomic data<br><br>Task 5. Predict ancestral genotypes of key bacterial species<br><br>Task 6. Perform in vitro antibiotic resistance assays<br><br>Task 7. Perform in vivo virulence assays using C. elegans<br><br>Task 8. ISS environmental sampling<br><br>Task 9. Estimate and identify culture-based microbial burden and diversity<br><br>Task 10. Perform metagenomic sequencing and qPCR of PMA-treated DNA to examine the viable phylogenetic and functional diversity |
| **Bibliography Type:** | Description: (Last Updated: 11/18/2025) |
| **Abstracts for Journals and Proceedings** | Minnis VR, Hill MS, Simpson AC, Venkateswaran K, Gilbert JA. "Antimicrobial resistance gene prevalence on the International Space Station. " UCSD Environmental Systems Symposium (ESYS) 2023, La Jolla, CA, September, 2023. Abstracts. UCSD Environmental Systems Symposium (ESYS) 2023, La Jolla, CA, September, 2023. Poster Session; Poster Award Winner. , Sep-2023 |
| **Abstracts for Journals and Proceedings** | Minnis VR, Hill MS, Simpson AC, Venkateswaran K, Gilbert JA. "Antimicrobial resistance gene prevalence on the International Space Station." AbGradCon23 (Astrobiology Graduate Conference 2023) Undergraduate Poster Session, La Jolla, CA, May 23, 2023.<br>Abstracts. AbGradCon23 (Astrobiology Graduate Conference 2023) Undergraduate Poster Session, La Jolla, CA, May 23, 2023. , May-2023 |
| **Abstracts for Journals and Proceedings** | Simpson AC, Hill MS, Urbaniak C, Gilbert JA, Venkateswaran K. "Microbial species distribution and bioburden aboard the ISS over time via the Microbial Tracking mission series." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022.<br>Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022 |
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| **Articles in Peer-reviewed Journals** | Simpson AC, Eedara VVR, Singh NK, Damle N, Parker CW, Karouia F, Mason CE, Venkateswaran K. "Comparative genomic analysis of Cohnella hashimotonis sp. nov. isolated from the International Space Station." Front Microbiol. 2023 Jun 15;14:1166013. https://doi.org/10.3389/fmicb.2023.1166013 ; PMID: 37396358; PMCID: PMC10308117 , Jun-2023 |
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